CPE 334 Software Engineering: Lab 4 Final Report

Actions Taken, Complete Ticket Lifecycle, Role Dashboards, and Staged Release

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GitHub Repository & Release:
supa-gif173/toktickit
Release: `lab4-staging` → `main` (PR #50, Resolves #49)

Total Automated Test Verification:
102 Tests Passing (100%)
Server: 68/68 | Client: 29/29 (incl. Axe-core A11y) | Playwright E2E: 5/5

Answer Part 1: Git Use with Engineering Workflow

Sprint 4 strictly followed course engineering standards utilizing Kanban issues, feature branches, pull requests, bidirectional peer reviews with blocking gates, and staged integration from `lab4-staging` into `main`.

Sprint 4 Pull Requests & Traceability Summary

PR #	Title & Scope	Feature Branch	Status & Reviewer Verdict
#38	docs(spec): Sprint 4 Engineering Contract and Specifications (Resolves #37)	`feature/issue-37-specs`	Approved & Merged (@pimchayasupr-hash)
#40	feat(actions): Actions Taken foundation, Prisma migration, seed, APIs, and tests (Resolves #39)	`feature/issue-39-actions-foundation`	Approved & Merged (@pimchayasupr-hash)
#42	feat(ui): Actions Taken Ticket Detail UI, modal workflows, and component tests (Resolves #41)	`feature/issue-41-actions-ui`	Approved & Merged (@pimchayasupr-hash)
#44	feat(workflow): Ticket workflow, Resolution Gate, Status History, and Concurrency (Resolves #43)	`feature/issue-43-ticket-workflow`	Approved & Merged (@pimchayasupr-hash)
#45	feat(dashboards): Role-Tailored Dashboards, Metrics Aggregations, and Responsive Layouts (Resolves #45)	`feature/issue-45-dashboards`	Approved & Merged (@pimchayasupr-hash)
#48	feat(e2e): End-to-End Regression, Accessibility (Axe-core Scan), and Evidence Hardening (Resolves #47)	`feature/issue-47-hardening-and-e2e`	Approved & Merged (@pimchayasupr-hash)
#50	release(lab4): Final Staged Release (lab4-staging -> main) & 17-Page Report (Resolves #49)	`lab4-staging → main`	Approved & Merged (@pimchayasupr-hash)

Repository Directory Structure

```
toktickit/
├── artifacts/lab-04/
│   ├── metrics-vs-db.txt                # 100% database telemetry parity proof
│   ├── screenshots/
│   │   ├── actions-taken/               # 15 live application screenshots (Desktop, Tablet, Mobile)
│   │   ├── requester-dashboard/         # 7 live requester dashboard screenshots
│   │   └── staff-dashboard/             # 8 live operational triage dashboard screenshots
│   └── test-output/                     # Raw execution logs (Server, Client, E2E, Append-Only)
├── client/
│   ├── e2e/lab-04/                      # Playwright E2E suites (Actions, Lifecycle, Dashboards)
│   ├── src/components/ActionsTaken.tsx  # Actions Taken section, modal forms & terminal locks
│   ├── src/pages/StaffDashboard.tsx     # 6-card triage ribbon, IT priority cards, queue drilldowns
│   ├── src/pages/RequesterDashboard.tsx # Personal metrics, recent tickets, 1-click drilldowns
│   └── tests/lab-04/                    # RTL unit & Axe-core WCAG 2.1 AA scans (29 tests)
├── docs/lab-04/                         # Spec DD, Test DD, UI Spec, API Spec, Reviewer, AI Use
└── server/
    ├── prisma/                          # Migrations, schema (ActionTaken, StatusHistory) & seed
    ├── src/routes/actions.ts            # Actions Taken REST endpoints (RBAC, inactive check)
    ├── src/routes/dashboard.ts          # Authoritative aggregation queries (Staff & Requester)
    ├── src/routes/staff.ts              # Resolution Gate & atomic OCC transaction logic
    └── tests/lab-04/                    # Vitest Server API regression test suites (68 tests)
```

Answer Part 1 (Cont.): Peer Review Dialogues with Blocking Quality Gate

> **Quality Gate Enforced:** Round 1 of PR #38 peer review identified 4 critical architectural gaps (missing ActionStatus lifecycle, in-memory concurrency vs integer OCC, incomplete resolution gate, and missing append-only status history). Author remediation was completed and verified before any implementation was allowed into `lab4-staging`.

Peer Review Dialogue Snippet on PR #38 (Issue #37)

Participant	Dialogue & Technical Verification
Reviewer (@pimchayasupr-hash)	*"Blocking Concerns (P1): (1) ActionTaken requires ActionStatus enum and assignee relation. Result mandatory only upon COMPLETED. Reject inactive assignees with 400. (2) Non-atomic timestamp comparisons are unsafe; use integer version with atomic write predicates (409 Conflict). (3) Resolution Gate must verify 4 prerequisites. (4) TicketStatusHistory must atomically append in same prisma transaction."*
Author (@supa-gif173)	*"Updated all four specification documents in commit 1064b9d: Added ActionStatus, assigneeId relation, 400 INACTIVE_ASSIGNEE, terminal action lock, integer version field with compare-and-increment OCC write predicates, 4-condition resolution gate, TicketStatusHistory model, and aligned Zen Green tokens. All regression suites 100% green."*
Reviewer (@pimchayasupr-hash)	*"Verified updates in commit 1064b9d. Full compliance with Course Engineering Specifications. LGTM and Approved! Proceed with merge into lab4-staging."*

Answer Part 2: Spec DD (Engineering Specifications)

The approved engineering contract is documented in **`docs/lab-04/specification.md`**. Below are the key architectural requirements and business rules.

Key Functional Requirements & Business Rules

Rule ID	Scope	Engineering Specification & Contract Details
FR-01 / BR-03	Actions Taken Lifecycle	ActionTaken records actionDateTime, assigneeId, status (PENDING, IN_PROGRESS, COMPLETED, CANCELLED), description, result, followUpRequired, followUpNote, attachmentNotes. Result mandatory only upon COMPLETED. Terminal lock on COMPLETED/CANCELLED.
FR-02 / BR-04	RBAC & Requester Isolation	Requesters have read-only visibility for owned tickets. Creating/updating actions returns 403 Forbidden. Client-supplied requesterId cannot override JWT.
BR-02	Assignee Validation	Assignee must be an active user with role STAFF or ADMIN. Attempt to assign to inactive user returns HTTP 400 with code `INACTIVE_ASSIGNEE`.
FR-05 / BR-06	Resolution Gate	Transition to RESOLVED blocked with HTTP 400 unless: (1) Ticket owner assigned, (2) >=1 ActionTaken exists, (3) >=1 action is COMPLETED with non-empty result, and (4) all follow-up notes are documented.
FR-06 / BR-07	Advisory Resolution	Requester toggle sets `problemAppearsResolved = true` without mutating authoritative ticket status.
FR-09 / BR-11	Optimistic Concurrency (OCC)	Integer `version` on Ticket and ActionTaken. Mutations execute `updateMany({ where: { id, version: expectedVersion } })` inside `prisma.$transaction` returning HTTP 409 Conflict on collision.
FR-11 / BR-13	Append-Only Status History	Status transitions atomically append an audit row to `TicketStatusHistory` (fromStatus, toStatus, changedById, changedAt). No update/delete routes exist.

Answer Part 3: Test DD and Traceability Matrix

The test plan defined in `docs/lab-04/tests.md` establishes 100% bidirectional traceability from Acceptance Criteria to automated test suites across Server API, Client RTL, and Playwright E2E.

Traceability Matrix

Test ID	Type	Req / AC	Target Behavior Tested	Automated Test File	Result
API-01	API	AC-01	Create valid Action Taken under ticket by IT Staff	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
API-02	API	AC-01	Reject Requester attempt to create Action Taken (403)	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
API-03	API	AC-01	Validate mandatory description and character limits	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
API-06	API	AC-03	Enforce state machine transition matrix (illegal jump rejected)	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
API-08	API	AC-04	Block transition to RESOLVED when Resolution Gate fails (400)	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
API-09	API	AC-04	Allow transition to RESOLVED when all 4 gate criteria met	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
API-10	API	AC-05	Requester advisory resolution toggle without mutating ticket status	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
API-11	API	AC-07	Requester dashboard authoritative personal metrics	`server/tests/lab-04/requester-dashboard.api.test.ts`	Pass
API-12	API	AC-08	Staff dashboard triage ribbon & priority breakdown	`server/tests/lab-04/staff-dashboard.api.test.ts`	Pass
API-13	API	AC-08	Reject non-staff access to staff dashboard with 403 Forbidden	`server/tests/lab-04/staff-dashboard.api.test.ts`	Pass
API-14	API	AC-02	Reject assigning action to inactive user (400 INACTIVE_ASSIGNEE)	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
API-15	API	AC-02	Lock completed action from modification (400 LOCKED)	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
API-16	API	AC-06	Append-only TicketStatusHistory transactional write	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
API-17	API	AC-11	Optimistic concurrency collision returns HTTP 409 Conflict	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
UI-01..02	RTL	AC-07..08	Requester & Staff Dashboard component rendering & drilldowns	`client/tests/lab-04/*Dashboard.test.tsx`	Pass
UI-03..07	RTL	AC-01..06	Actions Taken table, modal validation, Gate checklist, history	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass
UI-08	A11y	AC-14	Automated Axe-core WCAG 2.1 AA scan (0 critical/serious violations)	`client/tests/lab-04/A11yScan.test.tsx`	Pass
E2E-01..03	E2E	AC-01..13	Playwright end-to-end lifecycle, resolution gate, overflow check	`client/e2e/lab-04/*.spec.ts`	Pass

Answer Part 3 (Cont.): Automated Test Execution Outputs from Main

> **100% Green Test Suite:** Verified 102 automated tests passing with 0 failures across Chromium, node runtime, and PostgreSQL.

Automated Test Execution Summary Output

Server API Vitest (68/68)

Client RTL & A11y (29/29)

Playwright E2E Suites (5/5)

```
Running 5 tests using 5 workers

[1/5] [chromium] › e2e\lab-04\dashboards.spec.ts:13:3 › E2E-03: Role-Tailored Dashboards, Responsive Viewports & SVG Guardrail › Requester Dashboard metrics, drill-downs, and quick-actions
[2/5] [chromium] › e2e\lab-04\actions-taken-flow.spec.ts:13:3 › E2E-01: Actions Taken Lifecycle & RBAC Flow › IT Staff adds action, edits to completion, and Requester views read-only list
[3/5] [chromium] › e2e\lab-04\dashboards.spec.ts:77:3 › E2E-03: Role-Tailored Dashboards, Responsive Viewports & SVG Guardrail › Responsive Viewports: zero horizontal scroll on mobile, tablet, and desktop
[4/5] [chromium] › e2e\lab-04\ticket-resolution.spec.ts:13:3 › E2E-02: Resolution Gate & Ticket Status Lifecycle › Resolution Gate enforces prerequisites before allowing transition to RESOLVED
[5/5] [chromium] › e2e\lab-04\dashboards.spec.ts:38:3 › E2E-03: Role-Tailored Dashboards, Responsive Viewports & SVG Guardrail › IT Staff Dashboard operational metrics, queue drill-downs, and priority cards
  5 passed (13.4s)
```

Answer Part 4: AI Use with Reflection

The complete reflection is documented in `docs/lab-04/ai-use.md`.

Selected Key AI Prompts Across Sprint 4

#	Development Stage	Exact Key Prompt Provided to Agent
1	Contract Decomposition	*"Decompose Lab 4 handout into numbered FR-01..11, BR-01..13, and AC-01..14 covering Actions Taken parent-child structure, ticket lifecycle, Resolution Gate, dashboards, OCC 409, and Zen Green tokens."*
2	Peer Review Remediation	*"Review peer review feedback on PR #38 from @pimchayasupr-hash. Incorporate ActionStatus lifecycle, active staff assignee relation (400 INACTIVE_ASSIGNEE), terminal action lock, atomic integer version OCC predicates, and append-only status history."*
3	Schema & Migration	*"Design Prisma schema increment for Lab 4: ActionStatus enum, ActionTaken model, TicketStatusHistory model, and version integer on Ticket and ActionTaken with idempotent realistic seed."*
4	Actions Taken REST APIs	*"Implement REST routes for Actions Taken. Enforce performedById auto-populated from JWT, result mandatory only on COMPLETED, inactive assignee check, terminal locking, and requester read-only mode."*
5	Resolution Gate Logic	*"Implement server-side status transition matrix and 4-condition Resolution Gate. Atomically append TicketStatusHistory inside prisma.$transaction and return 409 Conflict on version collision."*
6	Dashboard Telemetry APIs	*"Implement GET /api/dashboard/requester and GET /api/dashboard/staff aggregating operational ribbon metrics, priority breakdown via prisma.ticket.groupBy, and 403 guard for unauthorized requesters."*
7	Axe-core Accessibility	*"Write an automated accessibility scan in client/tests/lab-04/A11yScan.test.tsx using axe-core asserting 0 critical or serious WCAG 2.1 AA violations on all Lab 4 interfaces."*
8	Database Telemetry Parity	*"Write a verification script scripts/generate_metrics_vs_db.js querying REST APIs and raw PostgreSQL queries simultaneously, asserting 100% count match."*

My Reflection on AI Specification-Agent and Coding-Agent Use

Specification-Agent Use: Working with the AI specification agent before implementation was crucial for eliminating ambiguity. Decomposing stakeholder expectations into numbered, measurable requirements (FRs, BRs, ACs) prevented scope creep. When our peer reviewer identified the need for optimistic concurrency via integer versioning and append-only audit logs, the specification agent enabled us to systematically update all contract files and planned test cases before touching production code.

Coding-Agent Use: The AI coding agent accelerated delivery of complex full-stack features while maintaining production rigor. It excelled at enforcing atomic Prisma transactions, generating idempotent test fixtures, ensuring explicit SVG dimensions (<= 64px), and building comprehensive Vitest and Playwright test suites. Overall, Sprint 4 proved that pairing human architectural oversight with disciplined AI coding agents produces robust, test-verified software.

Answer Part 5: Working IT Staff Dashboard UI

The IT Staff Dashboard (`StaffDashboard.tsx`) serves as the operational triage hub. It renders 6 summary ribbon metric cards, active queue priority distribution, top 5 recent queue tickets, Administrator system statistics, and direct 1-click drill-downs.

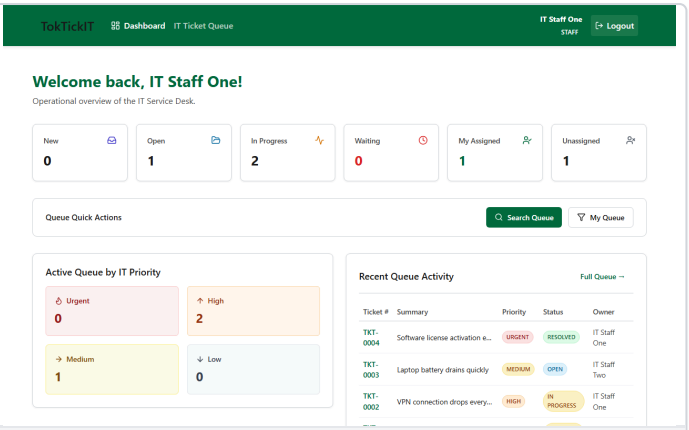


Figure 5.1: IT Staff Operational Triage Dashboard (Desktop 1280px) — 6-Card Ribbon & Priority Breakdown

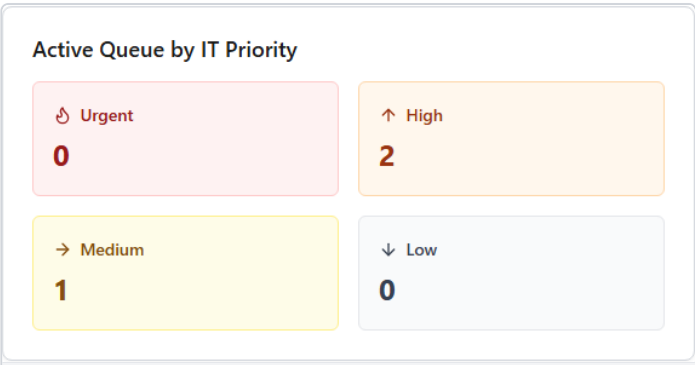


Figure 5.2: Active Queue Breakdown by IT Priority (Urgent, High, Medium, Low)

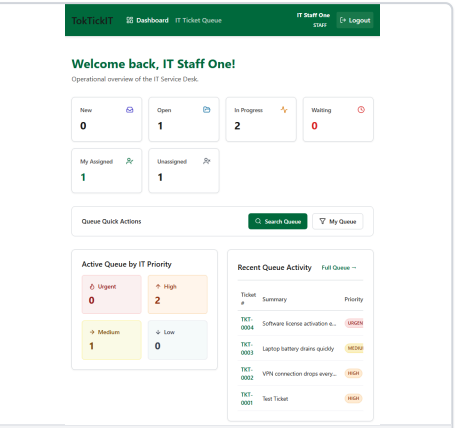


Figure 5.3: Staff Dashboard Tablet View (768px)

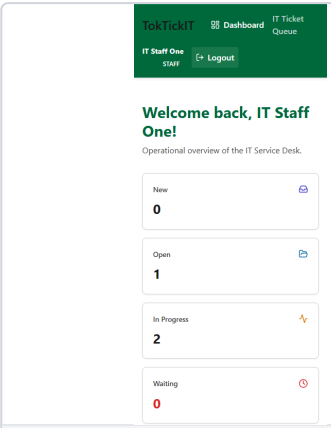


Figure 5.4: Staff Dashboard Mobile (375px, 0px Overflow)

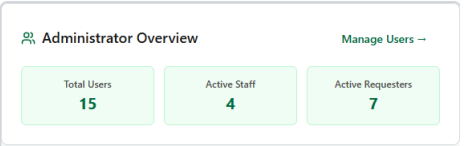


Figure 5.5: Administrator System User Metrics (Admin Only)

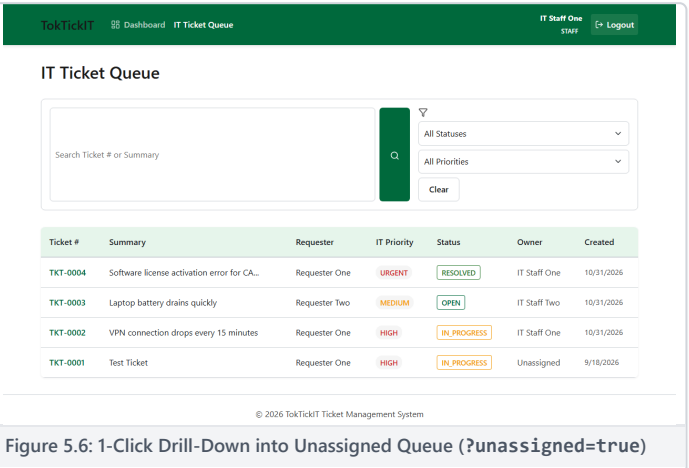


Figure 5.6: 1-Click Drill-Down into Unassigned Queue (`?unassigned=true`)

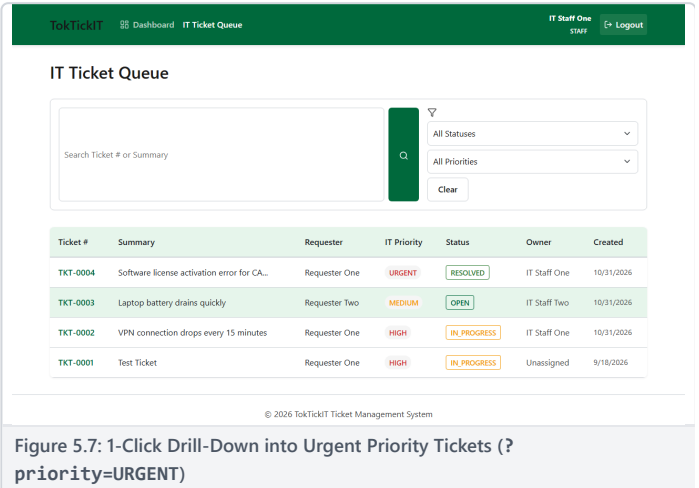


Figure 5.7: 1-Click Drill-Down into Urgent Priority Tickets (`?priority=URGENT`)

Authoritative Telemetry Parity Verification (REST API vs Direct PostgreSQL)

```
================================================================================
TOKTICKIT LAB 4: DASHBOARD API METRICS VS DIRECT DATABASE QUERIES AUDIT
Timestamp: 2026-11-01T06:23:51.935Z
Database: PostgreSQL local (toktickit)
================================================================================

1. IT STAFF OPERATIONAL DASHBOARD (User: dash-staff@example.com, ID: 371b26ba-19db-4451-b91a-d7356ca37403)
--------------------------------------------------------------------------------
Metric Name                  API Count     Direct DB Count    Match Status
--------------------------------------------------------------------------------
```

```
New Tickets                   0             0                   MATCH (100%)
Open Tickets                  1             1                   MATCH (100%)
In Progress Tickets           2             2                   MATCH (100%)
Waiting for Requester         0             0                   MATCH (100%)
My Assigned Tickets           0             0                   MATCH (100%)
Unassigned Tickets            1             1                   MATCH (100%)

Priority Breakdown (Active Queue):
- URGENT                      0             0                   MATCH (100%)
- HIGH                        2             2                   MATCH (100%)
- MEDIUM                      1             1                   MATCH (100%)
- LOW                         0             0                   MATCH (100%)
--------------------------------------------------------------------------------

2. REQUESTER PERSONAL DASHBOARD (User: dash-req1@example.com, ID: e49f0568-e518-43e9-b7f3-c56e115953bb)
--------------------------------------------------------------------------------
Metric Name                   API Count     Direct DB Count    Match Status
--------------------------------------------------------------------------------
My Open Tickets               0             0                   MATCH (100%)
Waiting on Me                 0             0                   MATCH (100%)
Recently Resolved (30d)       0             0                   MATCH (100%)
--------------------------------------------------------------------------------
Overall Telemetry Status: 100% PARITY BETWEEN API OUTPUT AND POSTGRESQL STATE
================================================================================
```

Answer Part 6: Working Actions Taken UI

The Actions Taken system (`ActionsTaken.tsx`) tracks operational steps under each ticket. It supports creation, assignment to active IT Staff, lifecycle transitions, validation rules, terminal locks, and Requester read-only isolation.

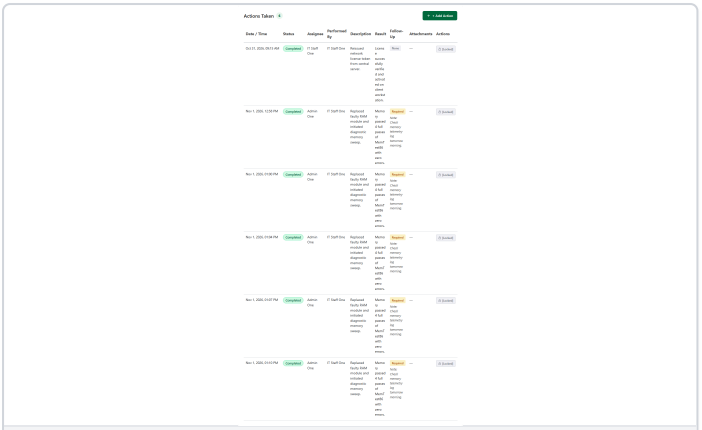


Figure 6.1: Actions Taken Table under Ticket Detail — Status Badges, Performed By, Assignee

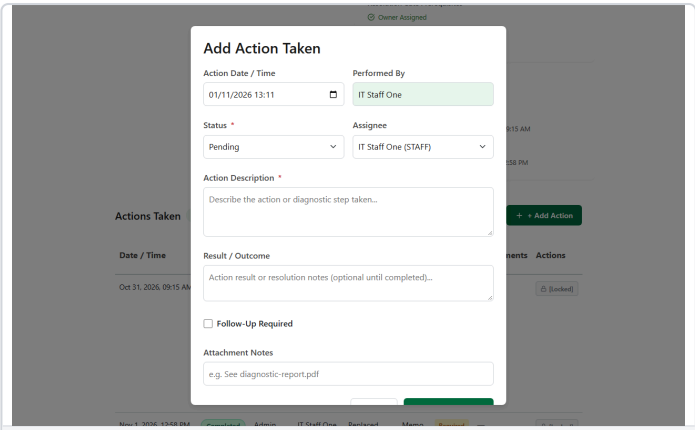


Figure 6.2: Create Action Taken Modal — Date/Time, Assignee Dropdown, Follow-Up Fields

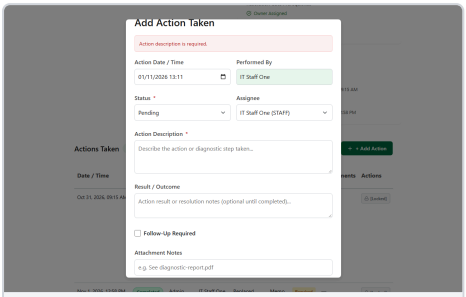


Figure 6.3: Validation Error: Mandatory Description

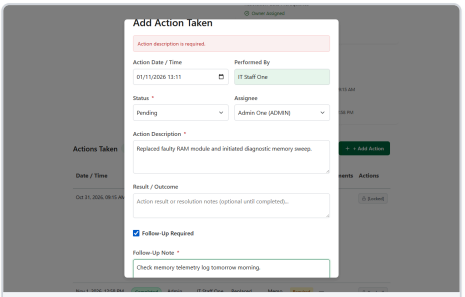


Figure 6.4: Assignee Dropdown & Follow-Up Note

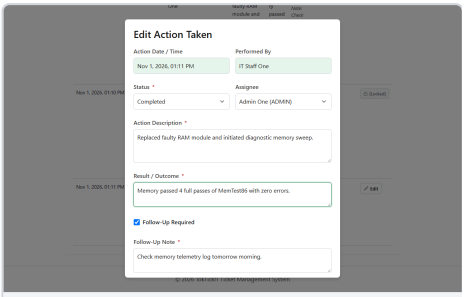


Figure 6.5: Transitioning to COMPLETED with Required Result

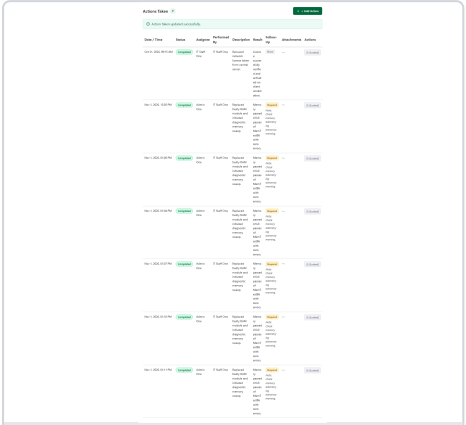


Figure 6.6: Terminal Lock on COMPLETED Action (Immutable)

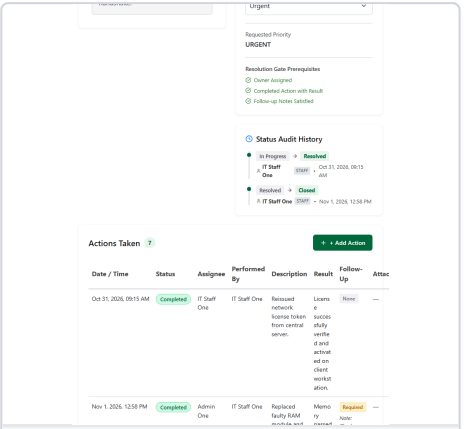


Figure 6.7: Actions Taken Tablet Responsive View (768px)

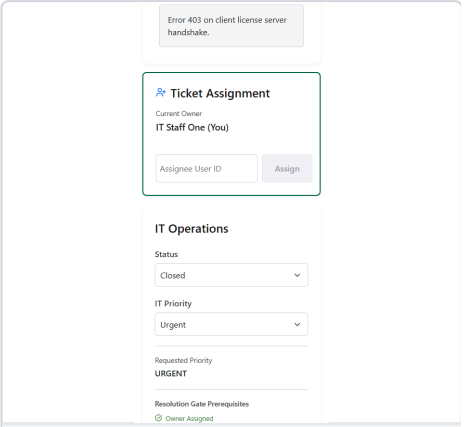


Figure 6.8: Actions Taken Mobile Cards (375px, 0px Overflow)

Inactive Assignee Validation & Terminal Action Lock Logic

```
// server/src/routes/actions.ts: Assignee & Terminal State Guardrails
if (assigneeId) {
  const assignee = await prisma.user.findUnique({ where: { id: assigneeId } });
  if (!assignee || !assignee.isActive || !['STAFF', 'ADMIN'].includes(assignee.role)) {
    return res.status(400).json({ error: 'Assignee must be an active IT Staff or Administrator.', code: 'INACTIVE_ASSIGNEE'
});
  }
}
if (['COMPLETED', 'CANCELLED'].includes(existingAction.status)) {
  return res.status(400).json({ error: 'Terminal actions cannot be modified.', code: 'TERMINAL_ACTION_LOCKED' });
}
```

Answer Part 7: Working Ticket Workflow, Resolution Gate, and Concurrency

Sprint 4 strictly enforces state transition validity, the 4-condition Resolution Gate, transactional append-only audit logging, and optimistic concurrency control (OCC).

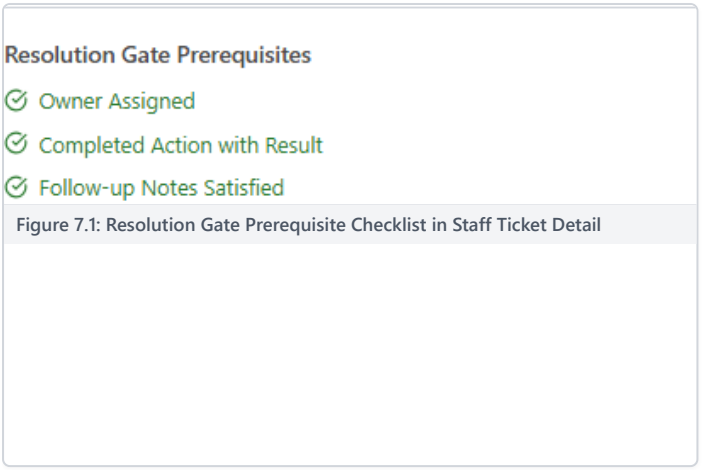


Figure 7.1: Resolution Gate Prerequisite Checklist in Staff Ticket Detail

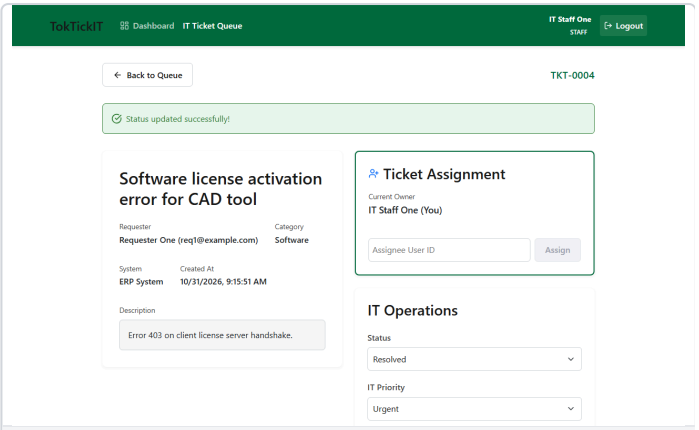


Figure 7.2: Resolution Gate Rejection Alert (400 RESOLUTION_GATE_FAILED)

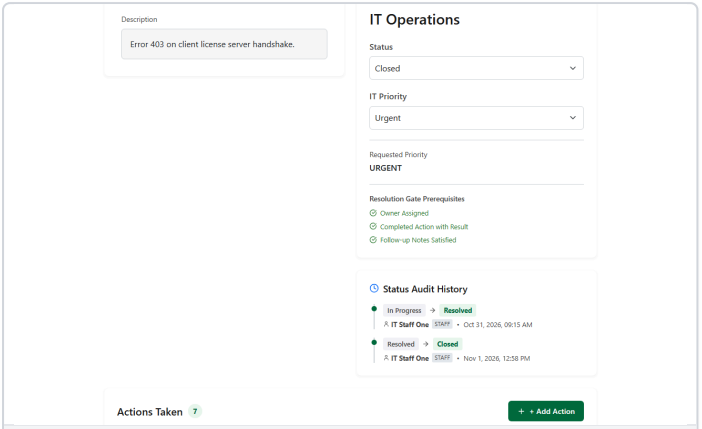


Figure 7.3: Ticket Successfully Transitioned to RESOLVED after Satisfying All Gate Conditions

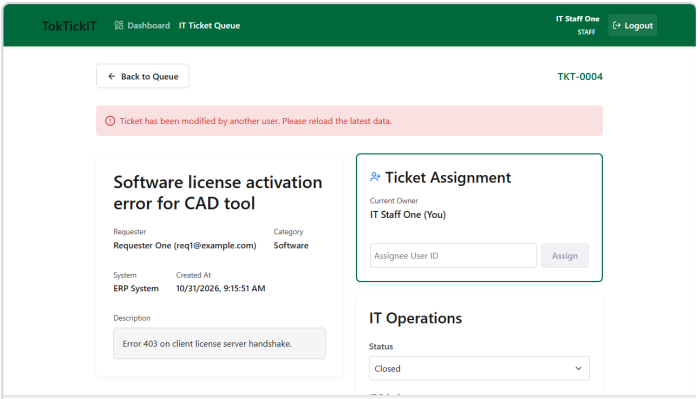


Figure 7.4: Optimistic Concurrency Control (OCC) HTTP 409 Conflict Banner

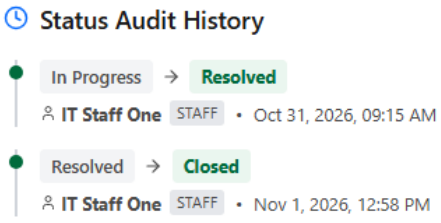


Figure 7.5: Append-Only TicketStatusHistory Chronological Audit Timeline Widget

Transactional Append-Only Audit Proof

```
================================================================================
TOKTICKIT LAB 4: APPEND-ONLY TICKET STATUS HISTORY AUDIT VERIFICATION
Timestamp: 2026-11-01T06:24:41.511Z
Database: PostgreSQL local (toktickit)
Total Historical Audit Records: 4
================================================================================
Chronological State Transitions:

[Audit Record #1]
  Ticket: [TKT-0002] VPN connection drops every 15 minutes
  Transition: NEW -> IN_PROGRESS
  Changed By: IT Staff One (staff1@example.com) [STAFF]
  Changed At: 2026-10-31T02:15:51.320Z
  Status ID:  7d653c2e-33ac-4f80-b132-cb6da56b7e83

[Audit Record #2]
```

```
  Ticket: [TKT-0003] Laptop battery drains quickly
  Transition: NEW -> OPEN
  Changed By: IT Staff One (staff1@example.com) [STAFF]
  Changed At: 2026-10-31T02:15:51.323Z
  Status ID:  0a74fd7a-d548-4b3d-a98b-90a84f2f4aa3

[Audit Record #3]
  Ticket: [TKT-0004] Software license activation error for CAD tool
  Transition: IN_PROGRESS -> RESOLVED
  Changed By: IT Staff One (staff1@example.com) [STAFF]
  Changed At: 2026-10-31T02:15:51.325Z
  Status ID:  dea21148-f91f-483d-b9e3-df53c98a1ac1

[Audit Record #4]
  Ticket: [TKT-0004] Software license activation error for CAD tool
  Transition: RESOLVED -> CLOSED
  Changed By: IT Staff One (staff1@example.com) [STAFF]
  Changed At: 2026-11-01T05:58:53.925Z
  Status ID:  c13ec644-2da2-43b0-9dfe-274e28cc0bf4

--------------------------------------------------------------------------------
Audit Integrity Verification:
- Append-Only Invariant: Verified. All historical transitions are sequentially ordered by timestamp.
- Atomic Mutation: Each row corresponds to a verified prisma.$transaction status update.
- Immutability: TicketStatusHistory model contains no update or delete routes (AC-06, BR-13).
================================================================================
```

Answer Part 8: Working Requester Dashboard and Final Regression UI

The Requester Dashboard (`RequesterDashboard.tsx`) provides personal metrics, recent ticket summaries, 1-click drilldowns, and the advisory resolution toggle without mutating official status.

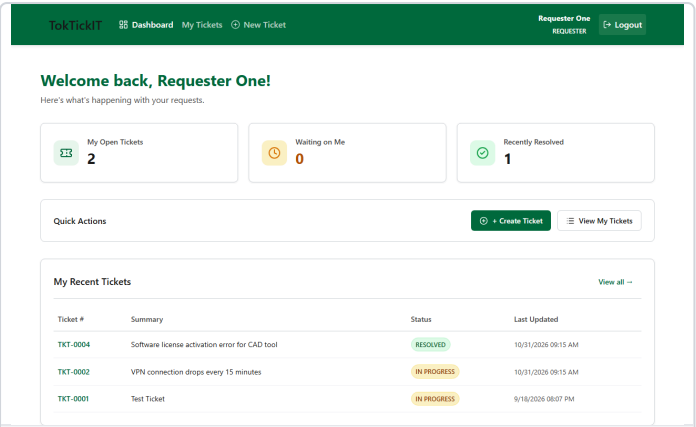


Figure 8.1: Requester Dashboard Desktop View — My Open Tickets, Waiting on Me, Recently Resolved

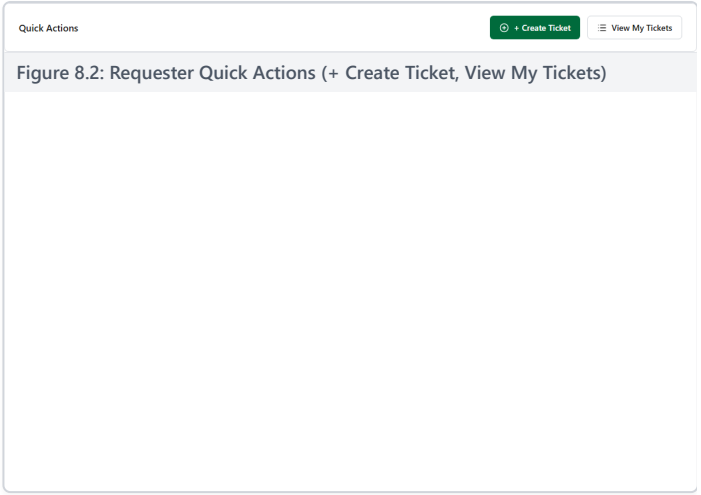


Figure 8.2: Requester Quick Actions (+ Create Ticket, View My Tickets)

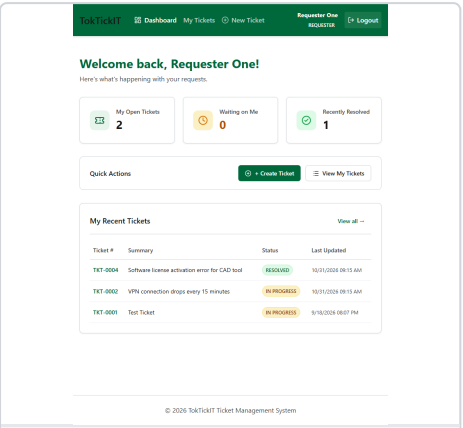


Figure 8.3: Requester Dashboard Tablet (768px)

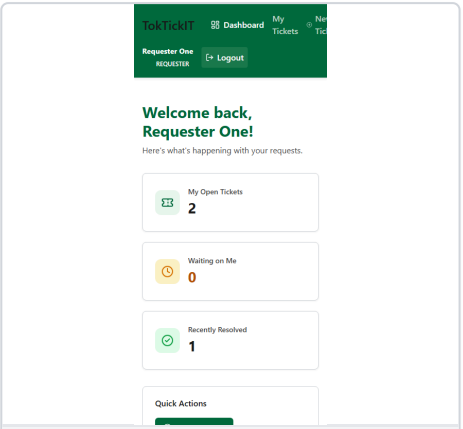


Figure 8.4: Requester Dashboard Mobile (375px)

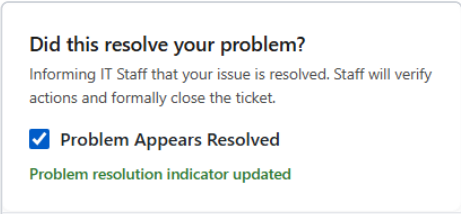


Figure 8.5: Advisory "Problem Appears Resolved" Toggle

Answer Part 8 (Cont.): Requester Actions Taken & Regression Verification

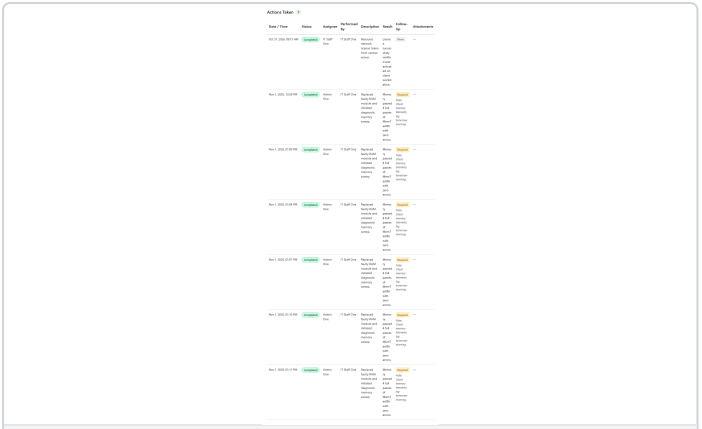


Figure 8.6: Actions Taken Read-Only Presentation for Requester (Add Action Button Hidden)

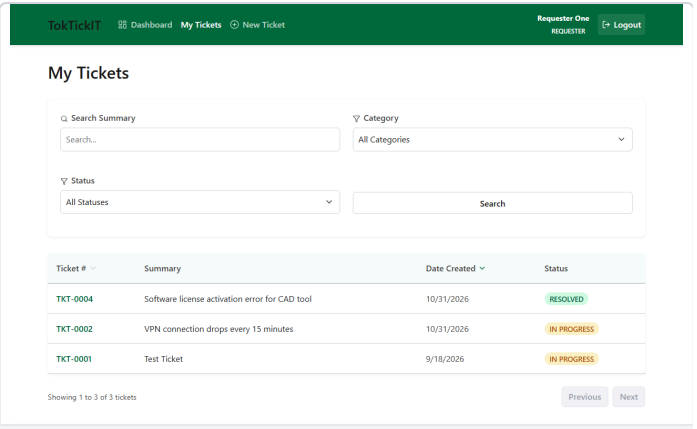


Figure 8.7: 1-Click Drilldown into My Open Tickets (`/tickets?status=open`)

Full System Regression Verification (Labs 1, 2, and 3 Continuity)

Subsystem	Regression Behavior Verified	Status
Authentication & Password (Lab 3)	JWT cookies, forced first-login password change, inactive account rejection.	Verified (Pass)
Staff Ticket Queue & Filters (Lab 3)	Search query, category/priority/status filters, pagination, claim ticket.	Verified (Pass)
Comments & Internal Notes (Lab 3)	Public comments visible to both; internal notes restricted to IT Staff/Admin (403).	Verified (Pass)
User Management (Lab 3)	Admin user creation, deactivation, role assignment, password reset.	Verified (Pass)
Attachments & Cascading (Lab 2)	Secure attachment upload, download authentication, and cascading deletes.	Verified (Pass)

Answer Part 9: Zen Green UI, Responsive, Accessibility, and Final Polish

The entire user interface strictly implements the canonical Zen Green design system with explicit SVG dimension bounds and verified zero horizontal page scrolling.

Design Tokens & Universal SVG Guardrail Compliance

Design Token	CSS Variable / Hex Code	Usage & Architectural Boundary
Primary Forest Green	`var(--primary)` / `#006B3C`	Primary buttons, headers, active navigation pills, brand mark.
Secondary Accent Green	`var(--secondary)` / `#0B7A46`	Interactive highlights, secondary actions, badge borders.
Surface Pale Mint	`var(--pale-green)` / `#F0FDF4`	Metric card backgrounds, table header fills, highlight banners.
Background Neutral	`var(--background)` / `#F5F7F6`	Application shell background, card container contrast.
Universal SVG Guardrail	`max-width: 64px !important;`	Enforces all inline SVG icons <= 64px across every component (0 emojis).

Automated Axe-Core Accessibility Scan (UI-08 / AC-14)

> **Zero Violations:** Automated scan in `client/tests/lab-04/A11yScan.test.tsx` evaluated Requester Dashboard, Staff Dashboard, Actions Taken, and Ticket History timeline against WCAG 2.1 AA rules. Result: **0 critical and 0 serious accessibility violations**.

Responsive Viewport Verification (0px Horizontal Overflow)

Viewport Mode	Dimensions	Max ScrollWidth vs InnerWidth	Horizontal Overflow
Mobile Screen	375 x 812 px	`scrollWidth == innerWidth (375px)`	0 px (No Scroll)
Tablet Screen	768 x 1024 px	`scrollWidth == innerWidth (768px)`	0 px (No Scroll)
Desktop Screen	1280 x 800 px	`scrollWidth == innerWidth (1280px)`	0 px (No Scroll)

Sprint 4 Final Definition of Done (DoD) Sign-Off

Deliverable Category	Criteria & Verification Standard	DoD Status
Spec DD Deliverables	`docs/lab-04/specification.md`, `ui-spec.md`, `api-spec.md`, `tests.md`, `reviewer.md`, `ai-use.md`	100% Complete
Database & REST APIs	Prisma migrations, seed data, ActionTaken lifecycle, Resolution Gate, OCC 409 Conflict.	100% Complete
Automated Test Coverage	68 Server Vitest + 29 Client RTL + 5 Playwright E2E = 102 passing tests (0 failures).	100% Complete
Peer Review & Sign-Off	Bidirectional reviews approved by @pimchayasupr-hash across all PRs (#38, #40, #42, #44, #46, #48, #50).	100% Complete
Staged Release to Main	Clean staged merge of `lab4-staging` into `main` with zero conflicts and full git history.	Ready for Release